

L6 – 1.5 - Solve Using the Q.F.

MCR3U

Part 1: DO IT NOW!

a) Do you remember the quadratic formula?

b) Use the quadratic formula to find the x-intercepts of:

$$0 = 2x^2 + 7x - 4$$

Don't forget that to solve a quadratic, it must be set equal to zero because at an x-intercept, the y-coordinate will be zero.

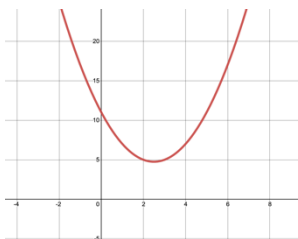
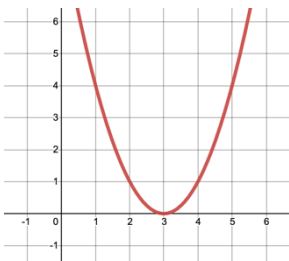
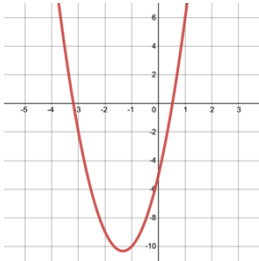
Part 2: Discriminant Review

Do all parabolas have two x-intercepts?

What are the three different scenarios?

The way to determine how many x-intercepts a parabola might have is by evaluating the $b^2 - 4ac$ part of the quadratic formula (called the "**discriminant**")

Discriminant: the value under the square root

# of solutions	0	1	2
Discriminant	$b^2 - 4ac < 0$	$b^2 - 4ac = 0$	$b^2 - 4ac > 0$
Example Equation	$0 = x^2 - 5x + 11$ $x = \frac{5 \pm \sqrt{(-5)^2 - 4(1)(11)}}{2(1)}$ $x = \frac{5 \pm \sqrt{-19}}{2}$ No real solutions Notice $b^2 - 4ac = -19$ which is less than zero	$0 = x^2 - 6x + 9$ $x = \frac{6 \pm \sqrt{(-6)^2 - 4(1)(9)}}{2(1)}$ $x = \frac{6 \pm \sqrt{0}}{2(1)}$ $x = 3$ Notice $b^2 - 4ac = 0$	$0 = 3x^2 + 8x - 5$ $x = \frac{-8 \pm \sqrt{(8)^2 - 4(3)(-5)}}{2(3)}$ $x = \frac{-8 \pm \sqrt{124}}{6}$ $x = -3.19, 0.52$ Notice $b^2 - 4ac = 124$ which is greater than zero
	 Notice the graph never crosses the x-axis	 Notice the vertex is ON the x-axis	 Notice the graph crosses through the x-axis twice

Another piece of useful information about the discriminant:

If $b^2 - 4ac$ is a _____ number, you will get rational solutions which means the quadratic is factorable. If you aren't sure if a quadratic is factorable, just check to see if $b^2 - 4ac$ is a perfect square number (0,1,4,9,16,25,36, ...)

Part 3: Solve a Quadratic with 2 Roots

Exact answer: as a radical or fraction. Exact answers do not have decimals.

Example 1: Find the exact solutions of $3x^2 - 10x + 5 = 0$

Example 2: Find the exact solutions of $-2x^2 + 8x - 5 = 0$

Part 4: Solving a Quadratic with 1 Root

***Note:** when a quadratic only has 1 solution, the x-intercept is also the vertex*

Example 3: Find the exact roots of $4x^2 + 24x + 36 = 0$

Part 5: Solving Quadratics with 0 Roots

2 Scenarios causing 0 roots:

- i) vertex is above the x-axis and opens up
- ii) vertex is below the x-axis and opens down

Example 4: Find the x-intercepts of $8x^2 - 11x + 5 = 0$

Part 6: Use the Discriminant to Determine the Number of Roots

Example 5: For each of the following quadratics, use the discriminant to state the number of roots it will have.

a) $2x^2 + 5x - 5 = 0$

b) $3x^2 - 7x + 5 = 0$

c) $-4x^2 + 12x - 9 = 0$

Part 7: Application

Example 6: A ball is thrown and the equation below model it's path:

$$h = -0.25d^2 + 2d + 1.5$$

' h ' is the height in meters above the ground and ' d ' is the horizontal distance in meters from the person who threw the ball.

a) At what height was the ball thrown from?

b) How far has the ball travelled horizontally when it lands on the ground?